Completion of the binary omission with intransitive subomissions

Human Compression project: a component of the L044 recovery

Let $G = A_n$ in its natural action, with distinguished generators $\rho_0, \dots, \rho_{r-1}$ forming a string C-marking. We acquire the exact finite maxima below degree fifteen, the complete transitive rank theorem, the completed pair-block theorem and the complete contracted-forest and neighbour components. The pair-block theorem's smaller-degree induction is stated explicitly in the result below.

Theorem 1. *Suppose $n \geq 12$, a transitive maximal omission $H = G_i$ preserves two blocks of size $m = n/2$, and every distinguished maximal omission $H_j = G_{i,j}$ is intransitive. Assume the alternating rank bound $\lfloor (q-1)/2 \rfloor$ in every smaller degree $12 \leq q < n$. Then*

$$r \leq \lfloor (n-1)/2 \rfloor.$$

Proof. The exact finite maxima settle $n \leq 14$. Suppose $n \geq 15$ and that the displayed bound fails, so $r \geq m$. Since H is transitive, some distinguished mark l interchanges its two blocks. It is a product of m transpositions, so evenness makes m even. Consequently $n \geq 16$ and $m \geq 8$.

Let C contain the other distinguished marks of H commuting with l, and let R contain the remaining marks. String relations give $|R| \leq 2$. First suppose R is empty. Choose coordinates matching the two rows through l. Then $H \leq \mathrm{diag}(S_m) \times \langle l \rangle$, and the projection of $\langle C \rangle$ on the l-pairs is faithful: its kernel is contained in $\langle l \rangle$, while the actual intersection property gives $\langle C \rangle \cap \langle l \rangle = 1$. This image P is transitive on the pairs, because H is transitive and adjoining l changes no pair index. It has the actual inherited string C-marking of rank $r - 2 \geq m - 2$.

For $m \geq 10$, the complete transitive rank theorem forces $P = S_m$ or A_m, since $m - 2 > m/2 + 1$. At $m = 8$, the complete proper transitive boundary has no rank-six exception, giving the same conclusion. Thus the pair action is primitive and the l-pair system is maximal. The completed pair-block theorem for the parent A_{2m} excludes it under the present explicit smaller-degree induction. This establishes maximality before that theorem is invoked; preservation of pairs alone would not have sufficed.

Now suppose $R \neq \varnothing$. Select one actual edge for each mark of H other than l, joining different orbits of its actual omission, and contract the l-pairs. The complete forest theorem gives $r \leq m$: a noncommuting neighbour has at least two distinct contracted edges, which cannot both occur within a connected selected forest. Thus the assumed failure forces $r = m$, with exactly two selected components.

If $|R| = 1$, the complete one-neighbour theorem excludes the case where the neighbour preserves the binary rows. If it interchanges them, that theorem changes the distinguished block swap to this neighbour and gives exactly two noncommuting neighbours, with the same H, omitted label and binary partition. All other standing hypotheses are unchanged. Thus it remains only to treat $|R| = 2$.

Write these two neighbours as u, v. The complete common bridge theorem gives rank three for their full contracted graph. The four-pair reduction makes this graph connected on exactly four pairs. If u, v both preserve the rows, the complete preserving neighbour theorem excludes the configuration by its actual classification, subgroup intersections and endpoint centralizers. If at least one interchanges the rows, the complete swapping neighbour theorem constructs a central pair system, proves it maximal, and applies the completed pair-block theorem. Its only alternating induction premise is the present bound in strictly smaller degrees. The one-neighbour

reduction is used only to enter this two-neighbour case; neither two-neighbour component invokes a completed one-neighbour exclusion, so this routing is not circular.

Every possibility for R is excluded. Hence a rank failure is impossible, proving the theorem. □

Attribution and scope. This completes the branch corresponding to source Proposition 3.5 of Cameron–Fernandes–Leemans–Mixer, conditional on the stated smaller-degree induction. The eligible-edge strategy is borrowed; the actual support, attachment, local permutation and centralizer arguments are retained in full in their component manuscripts. The empty-R maximality interface and this assembly were supplied by the continuation investigator. The distinct authorship and review contributions to each component remain in their records. This is an assembly of the fixed principal-source contract, not a new conjecture claim or a whole-source disposition.